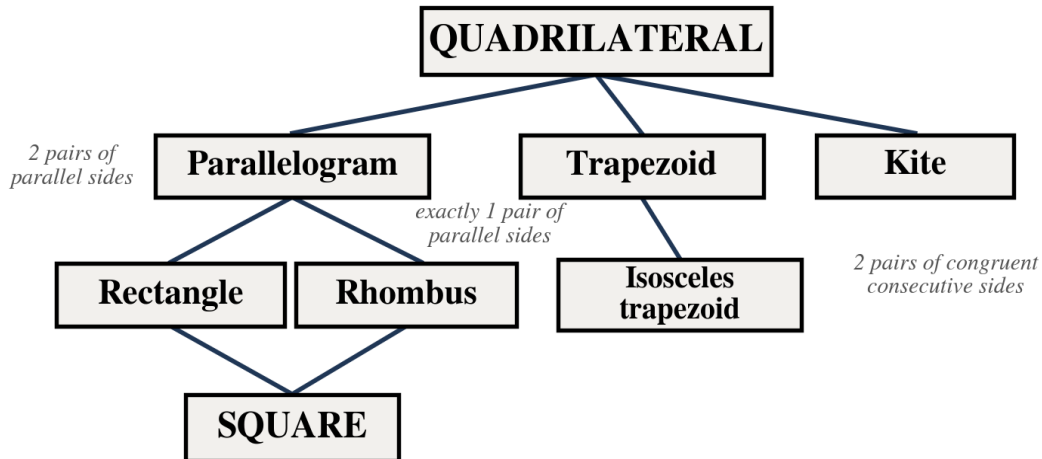


The Quadrilateral Family — A Reference Guide

One family tree, six shapes. This page organizes them; the next page is the angle arithmetic; the last is practice.



Shape	Definition	What you get for free
Parallelogram	Both pairs of opposite sides parallel.	opp. sides $\cong$ · opp. angles $\cong$ · consecutive angles supplementary · diagonals bisect each other
Rectangle	Four right angles.	everything a parallelogram has, PLUS congruent diagonals
Rhombus	Four congruent sides.	everything a parallelogram has, PLUS perpendicular diagonals that bisect the corner angles
Square	Four right angles AND four congruent sides.	everything a rectangle has AND everything a rhombus has
Trapezoid	Exactly one pair of parallel sides (the bases).	isosceles version: base angles $\cong$ and diagonals $\cong$ · midsegment = half the sum of the bases
Kite	Exactly two pairs of congruent CONSECUTIVE sides.	perpendicular diagonals · exactly one pair of congruent opposite angles

Reading the Tree — Inheritance Is the Whole Game

- Every arrow means IS-A. A rectangle IS a parallelogram, so every parallelogram fact is automatically a rectangle fact. You never re-prove inherited properties.
- The square sits at the bottom because it inherits from BOTH branches: it is a rectangle (congruent diagonals) and a rhombus (perpendicular, angle-bisecting diagonals) at the same time.
- Trapezoids and kites are NOT parallelograms. A trapezoid has exactly ONE pair of parallel sides; a kite has NO parallel sides — so none of the parallelogram properties apply to them.
- 'Always / sometimes / never' questions are tree questions. A rhombus is SOMETIMES a rectangle (when it is a square). A square is ALWAYS a rhombus. A trapezoid is NEVER a parallelogram.

The Angle Arithmetic You Actually Use

Almost every Chapter 7 computation is one of these five moves.

- Interior sum of an n-gon: $(n - 2) \cdot 180^\circ$. Each angle of a REGULAR n-gon: $(n - 2) \cdot 180^\circ \div n$.
- Exterior angles of ANY convex polygon total 360° , one per vertex. Each exterior angle of a regular n-gon: $360^\circ \div n$.
- Parallelogram angles: opposite angles are equal, consecutive angles add to 180° . One angle tells you all four.
- Diagonal work: in a parallelogram the diagonals BISECT each other (set the halves equal). In a rectangle they are also CONGRUENT. In a rhombus they are also PERPENDICULAR and bisect the corner angles.
- Trapezoid midsegment: half the SUM of the bases — not the average of two random sides, the two BASES.

Which Quadrilateral Am I? — Ask in This Order

- 1. How many pairs of parallel sides? Two → the parallelogram branch. Exactly one → trapezoid. None (but two pairs of congruent consecutive sides) → kite.
- 2. On the parallelogram branch: right angles? all sides congruent? Both → square. Only right angles → rectangle. Only congruent sides → rhombus. Neither → plain parallelogram.
- 3. Check with the diagonals if sides and angles are hidden: congruent → rectangle-ish · perpendicular → rhombus-ish · both → square.

Name: _____

Sorting Practice

Name the most specific quadrilateral that **MUST** fit each description — or write NOT ENOUGH if more than one branch survives.

- 1.** Both pairs of opposite sides are parallel.
- 2.** Four congruent sides.
- 3.** Four right angles.
- 4.** Four right angles and four congruent sides.
- 5.** Exactly one pair of parallel sides.
- 6.** Exactly two pairs of congruent consecutive sides.
- 7.** A parallelogram whose diagonals are congruent.
- 8.** A parallelogram whose diagonals are perpendicular.
- 9.** A parallelogram whose diagonals are congruent AND perpendicular.
- 10.** A trapezoid whose diagonals are congruent.
- 11.** A quadrilateral whose diagonals are perpendicular.
- 12.** A quadrilateral whose diagonals bisect each other.
- 13.** A rhombus with one right angle.
- 14.** A quadrilateral with exactly one pair of congruent opposite angles.
- 15.** A parallelogram with one pair of consecutive sides congruent.